

EDSON, AB, Canada

WMO: 711670

Lat: 53.5805N

Lon: 116.4535W

Elev: 927

StdP: 90.67

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.8	-28.0	-36.4	0.1	-31.5	-32.2	0.2	-27.5	9.6	3.3	8.0	2.1	1.8	280	0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	27.1	16.0	25.2	15.2	23.6	14.5	17.7	24.5	16.6	22.9	15.5	21.5	3.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.2	12.1	20.2	14.1	11.2	18.8	13.1	10.5	17.6	53.7	24.7	49.9	22.9	46.6	21.5	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 7.9	6.6	5.5		-36.7	30.6	3.8	1.5	-39.4	31.6	-41.6	32.5	-43.7	33.3	-46.5		
(5)				-36.7	19.7	3.6	1.2	-39.3	20.6	-41.4	21.3	-43.4	22.0	-46.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-10.4	-7.9	-3.8	3.2	8.8	12.8	15.6	14.3	9.6	3.1	-5.1	-10.5		
	DBStd	10.96	8.84	7.50	7.29	4.60	4.15	2.98	2.99	3.15	3.99	4.48	6.49	7.63		
	HDD10.0	3213	633	501	429	207	72	1	4	54	217	453	636			
	HDD18.3	5783	891	734	687	454	296	167	94	128	262	473	703	894		
	CDD10.0	486	0	0	0	2	34	92	174	138	43	3	0	0		
	CDD18.3	14	0	0	0	0	0	2	9	3	0	0	0	0		
	CDH23.3	414	0	0	0	1	28	54	188	111	31	2	0	0		
	CDH26.7	67	0	0	0	0	3	7	36	17	3	0	0	0		
(14) Wind	WSAvg	2.2	2.0	2.1	2.3	2.6	2.7	2.5	2.4	2.1	2.1	2.1	1.9	2.1		
(15) Precipitation	PrecAvg	597	28	17	24	24	60	115	115	84	67	21	21	21		
	PrecMax	825	82	51	50	61	180	254	228	192	153	45	45	47		
	PrecMin	423	2	0	6	3	20	44	38	12	17	5	4	3		
	PrecStd	129	23	11	13	15	36	51	59	52	39	10	14	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	10.4	13.8	21.3	26.6	27.6	30.1	28.7	26.8	20.9	12.0	8.6		
		MCWB	3.6	3.4	5.8	9.6	12.4	15.3	17.2	17.4	14.9	11.1	5.7	2.9		
	2%	DB	6.4	7.1	10.5	17.4	23.1	24.8	27.4	26.1	23.7	16.3	7.7	5.2		
		MCWB	1.8	1.8	3.7	8.0	11.6	14.0	16.7	16.3	13.9	8.5	3.2	0.9		
	5%	DB	3.8	4.9	8.3	14.6	20.8	22.8	25.4	24.2	21.1	13.2	5.2	2.3		
		MCWB	0.3	0.4	2.4	6.3	10.7	13.2	15.9	15.7	13.0	6.9	1.6	-0.7		
	10%	DB	1.0	2.5	6.1	11.9	18.3	20.7	23.5	22.3	18.1	10.6	2.8	-0.1		
		MCWB	-1.1	-0.9	1.1	4.8	9.6	12.4	15.1	14.9	11.7	5.6	0.2	-2.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	3.9	6.1	10.2	14.1	16.6	19.3	19.1	16.4	11.8	6.0	3.3		
		MCDB	8.5	9.3	13.3	20.0	22.5	23.6	26.1	25.6	23.5	19.7	10.4	7.7		
	2%	WB	2.2	2.2	4.1	8.5	12.6	15.4	18.0	17.7	14.7	9.4	3.7	1.3		
		MCDB	6.0	6.7	9.7	16.5	20.9	22.3	25.2	24.1	21.7	15.1	7.0	4.7		
	5%	WB	0.6	0.7	2.7	6.9	11.4	14.2	16.9	16.6	13.4	7.6	2.0	-0.5		
		MCDB	3.2	4.4	7.6	13.4	19.0	20.4	23.4	22.3	19.8	12.3	4.7	2.0		
	10%	WB	-1.2	-0.7	1.5	5.3	10.2	13.2	15.7	15.4	12.0	6.1	0.4	-2.3		
		MCDB	1.1	2.3	5.6	11.0	17.3	18.9	21.8	21.0	17.7	9.9	2.3	-0.2		
(35) Mean Daily Temperature Range	MDBR		11.9	13.4	12.5	12.7	13.4	12.5	12.8	13.1	13.3	11.7	10.1	10.9		
	5% DB	MCDBR	14.1	16.0	15.6	18.4	19.0	17.2	16.9	17.3	19.8	16.8	12.9	12.6		
		MCWBR	10.2	11.2	9.7	10.3	9.2	8.2	7.8	8.7	10.6	10.1	8.9	9.5		
	5% WB	MCDBR	13.5	15.2	14.7	17.0	16.7	14.4	14.9	15.3	17.7	15.6	12.1	12.4		
		MCWBR	10.2	10.8	9.4	9.8	8.5	7.2	7.3	8.3	10.0	9.8	8.8	9.7		
(40) Clear-Sky Solar Irradiance	taub		0.249	0.258	0.282	0.324	0.339	0.354	0.352	0.356	0.315	0.281	0.255	0.237		
	taud		2.487	2.561	2.531	2.430	2.431	2.400	2.458	2.411	2.538	2.609	2.536	2.439		
	Ebn at Noon		772	884	920	910	910	895	891	865	868	839	759	712		
	Edh at Noon		48	62	79	101	106	111	103	102	79	59	46	41		
(44) All-Sky Solar Radiation	RadAvg		0.89	1.80	3.11	4.35	5.11	5.44	5.60	4.74	3.39	1.96	0.97	0.67		
	RadStd		0.05	0.13	0.21	0.40	0.43	0.46	0.41	0.33	0.36	0.19	0.07	0.05		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page